

YAMBA, Australia

WMO: 945890

Lat: 29.4333S

Lon: 153.3633E

Elev: 92

StdP: 14.65

Time Zone: 10.00 (AUS)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.8	48.4	31.6	26.0	58.4	35.4	30.4	57.4	24.7	64.4	20.3	64.4	7.4	240	0.380

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	84.7	75.1	82.8	74.1	81.4	73.3	78.0	81.2	76.7	80.2	75.6	78.9	9.1	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	141.9	79.8	75.7	134.9	78.4	74.5	129.7	77.6	41.6	80.9	40.3	80.5	39.2	79.0	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 23.2	20.2	17.9	DB	43.1	92.6	1.1	4.1	42.3	95.6	41.6	98.0	41.0	100.4	40.2	103.4	(4)
(5)			WB	38.9	79.2	1.2	2.8	38.0	81.2	37.3	82.8	36.7	84.4	35.8	86.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.8	75.3	75.3	74.0	69.6	64.1	60.3	59.0	60.5	64.7	67.6	70.9	73.1
	DBStd	6.64	2.57	2.72	2.66	3.10	3.42	3.32	3.58	3.91	3.87	3.74	3.09	2.80
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	614	0	0	0	3	58	142	188	151	53	18	1	0
	CDD50	6503	785	708	743	587	436	310	281	324	440	545	626	718
	CDD65	1642	320	288	278	140	29	2	4	10	43	98	177	253
	CDH74	6808	1698	1595	1215	355	63	2	9	57	108	251	491	964
	CDH80	705	192	185	110	24	1	0	0	13	12	27	54	87
Wind	WSAvg	8.7	9.1	10.1	9.2	9.0	7.6	8.1	7.7	8.2	8.6	9.2	9.1	9.0
Precipitation	PrecAvg	60.8	5.3	6.8	8.0	6.9	7.2	5.4	3.4	2.4	1.9	3.3	4.0	5.1
	PrecMax	96.0	18.9	22.0	24.3	24.8	21.8	21.6	11.8	7.2	8.4	14.6	9.9	15.2
	PrecMin	32.5	0.5	1.5	1.6	1.2	0.4	0.5	0.1	0.1	0.0	0.2	0.2	1.5
	PrecStd	15.3	3.7	4.0	5.3	5.0	5.1	3.9	3.0	2.0	1.9	2.7	2.3	3.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.7	86.6	85.9	82.8	79.1	73.0	74.9	79.8	80.8	83.0	86.7	85.4
		MCWB	76.4	78.0	78.1	72.6	68.8	64.1	62.6	63.3	64.8	67.9	66.7	73.5
	2%	DB	83.7	84.1	82.7	79.4	75.5	70.2	71.3	74.0	76.6	78.9	80.4	81.9
		MCWB	75.4	75.7	75.5	71.7	66.4	62.1	61.5	61.2	65.0	68.6	71.7	72.6
	5%	DB	82.0	82.1	80.7	77.4	72.9	68.4	68.7	70.9	74.3	76.5	78.4	80.2
		MCWB	74.3	74.1	73.9	70.1	65.3	61.5	60.7	60.6	64.8	68.0	70.4	72.1
	10%	DB	80.5	80.6	79.1	75.8	71.1	67.0	66.5	68.4	72.0	74.6	76.6	78.7
		MCWB	73.7	73.2	72.6	69.1	64.2	60.9	59.6	59.4	63.8	67.1	69.4	71.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	80.1	79.6	76.2	72.1	68.0	67.7	67.1	70.2	73.4	74.7	77.2
		MCDB	82.2	83.0	83.7	78.2	75.8	69.3	70.8	74.1	75.1	77.0	80.0	81.6
	2%	WB	77.2	78.2	77.4	73.7	69.7	65.5	65.3	65.1	68.5	71.4	73.2	75.2
		MCDB	81.1	81.0	80.3	77.1	72.4	67.8	68.3	69.8	73.0	75.4	78.0	78.8
	5%	WB	76.1	76.7	76.0	71.9	67.6	64.0	62.9	63.5	67.2	69.8	72.0	73.9
		MCDB	79.8	80.1	79.0	76.2	70.8	66.6	66.5	68.2	71.4	74.0	76.1	77.7
	10%	WB	75.0	75.2	74.6	70.2	65.9	62.6	61.0	61.8	65.6	68.4	71.0	72.9
		MCDB	78.4	78.6	77.7	74.4	69.7	65.4	64.8	66.7	70.2	72.6	74.9	76.7
Mean Daily Temperature Range	5% DB	MDBR	10.2	10.1	10.5	11.9	14.3	13.5	15.4	16.8	14.7	12.8	11.1	10.6
		MCDBR	12.4	12.2	12.6	14.3	16.6	15.0	17.7	20.0	17.3	16.4	13.8	13.2
		MCWBR	5.9	5.7	6.7	7.9	9.5	9.3	10.8	11.7	10.3	9.0	7.3	6.8
	5% WB	MCDBR	10.8	11.0	11.8	13.5	14.1	12.0	14.2	17.2	15.1	14.0	11.5	11.7
		MCWBR	6.0	6.0	7.3	8.2	9.1	9.1	9.7	11.5	9.8	8.7	6.8	6.6
Clear-Sky Solar Irradiance	taub	0.388	0.373	0.357	0.339	0.315	0.310	0.301	0.314	0.356	0.385	0.385	0.392	
	taud	2.512	2.565	2.605	2.630	2.655	2.655	2.666	2.610	2.488	2.436	2.484	2.485	
	Ebn at Noon	303	301	297	287	280	274	282	291	292	294	301	302	
	Edh at Noon	36	33	30	27	23	22	23	27	33	37	37	37	
All-Sky Solar Radiation	RadAvg	2081	1901	1592	1346	1098	910	1051	1337	1676	1876	2027	2068	
	RadStd	165	145	91	67	68	55	82	93	118	161	170	191	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		+0.61	N/A	N/A	+1.15	N/A	N/A	N/A	-81	+290	+182

Nomenclature: See separate page